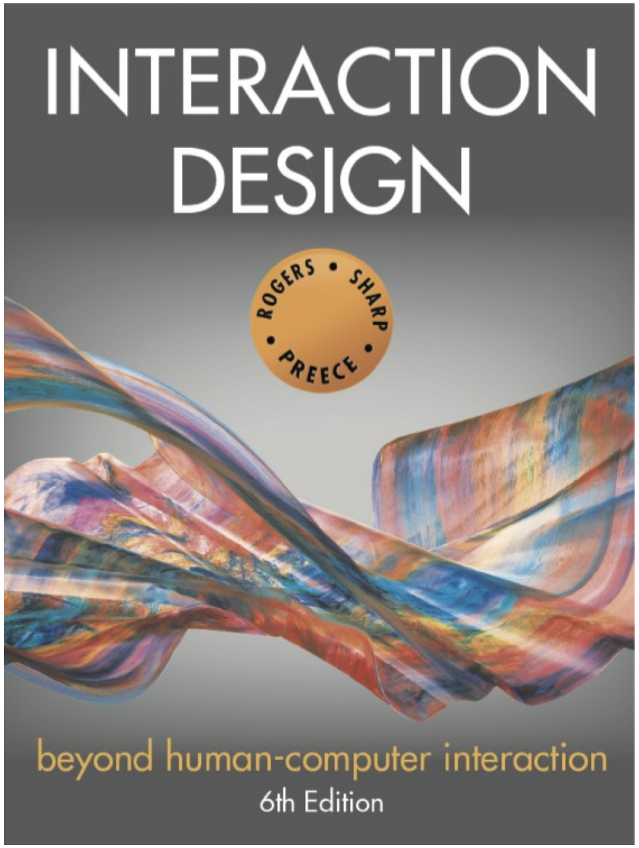


Chapter 5
SOCIAL INTERACTION

Overview

- What is meant by social interaction
- The social mechanisms used in conversations
- What is meant by social presence
- Overview of technologies for supporting social interaction, social presence and social games
- Illustrate how video conferencing and other social media platforms evolved during COVID-19
- How technology can augment people who are co-present

Social interaction

- We live together, work together, play together, talk to each other, and socialize
- We will focus on:
  - how people communicate and collaborate face-to-face and remotely in their social, work, and everyday lives
  - providing deeper understandings and guidance for the design of social technologies that can better support and extend them

Social interaction

- The use of social media has dramatically increased.
- People now spend several hours a day communicating with others online:
  - messaging, emailing, tweeting, videoconferencing, and so on.
- It is also common practice for people at work to keep in touch with each other via
  - WhatsApp/WeChat groups and other workplace communication tools, such as Slack, Yammer, or Teams.
- Widespread adoption of shared calendars and collaboration tools (e.g., Miro and Google Docs)

Social interaction

- COVID-19
  - During the COVID pandemic, there was a rapid and widespread uptake of videoconferencing software, notably Teams and Zoom, that enabled people to keep in touch with each other, study, and continue working.
  - Within a matter of a few weeks, the way people worked or studied was transformed.

Social Interaction

Being Social during Covid-19

Online zoom meetings often accompanied by eating and drinking separately but together while on the same video call.

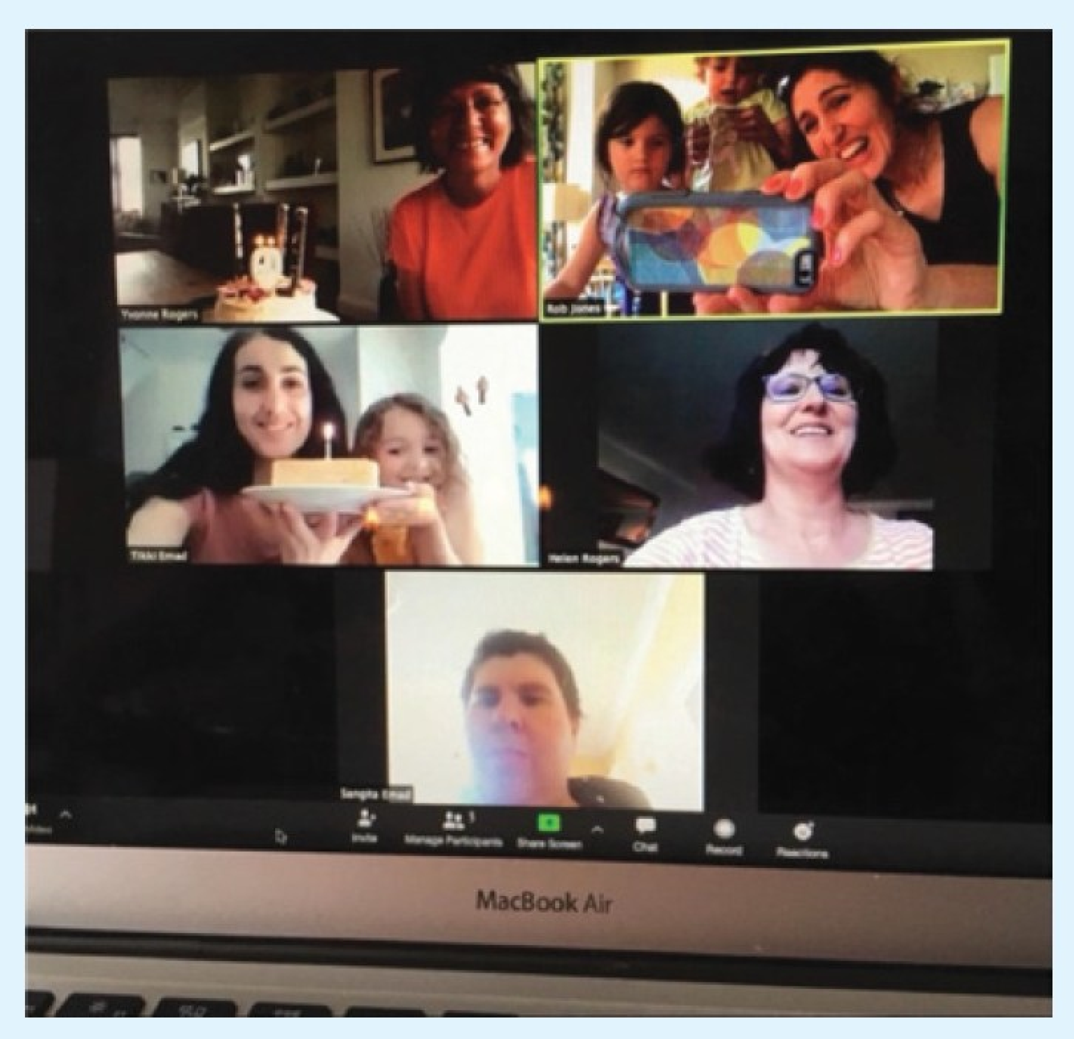


Figure: A Zoom birthday party in full swing

Social Interaction

- **A growing concern** - how much time people spend looking at their phones
  - when interacting with others, playing games, tweeting, and so forth
  - and its consequences on people's well-being.
- Findings of a survey conducted in the United States in 2021:
  - more than half the people who replied said they spend **five to six hours** on their phone on a daily basis,
  - and that is not including work-related smartphone use (Ceci, 2022).
  - Similar amounts were found for users in Brazil, Indonesia, and South Korea

Social Interaction

- Social technologies developed to enable us to persist in being social when apart
  - They differ in how they support us
    - **Positive effect**: Some encourage social interactions (for example, family games with Amazon's virtual assistant, Alexa)
    - **Negative effect**: Others have a negative impact on everyday life, especially in-person conversations (Sherry Turkle, 2015)

Are we spending too much time in our own digital bubbles?

Fig. A family sits together, but they are all in their own **digital bubbles**—including the dog

Social Interaction in the age of social media

- New social rules have emerged for how to behave and interact with others when communicating via videoconferencing. For example,
  - muting yourself when not talking.
  - when wanting to speak, raise your hand virtually by clicking the hand icon.
  - emoji reactions to signify various forms of praise, emotions, and nonverbal feedback
- In the past, people would usually call each other to arrange to meetup,
  - many now text each other or create a WhatsApp group if it is a group event.

Questions raised by social tech

- Are in person conversations being superseded by social media interactions?
- How many friends do you have on Facebook, LinkedIn, WhatsApp and so on versus real life?
- How much do they overlap?
- How are the ways that we live and interact with one another changing?
- Are the established rules and etiquette still applicable to online and offline?

Conversational mechanisms

Various mechanisms and ‘rules’ are followed when holding a conversation face to face, such as mutual greetings

A: Hi there
B: Hi!
C: Hi
A: All right?
C: Good, how’s it going?
A: Fine, how are you?
C: OK
B: So-so. How’s life treating you?

Conversational mechanisms

- Understanding how conversations start, progress, and finish is useful when designing dialogues that take place with chatbots, voice assistants, and other communication tools

Conversational rules

Sacks et al. (1978) conversation analysis of conversations propose three basic rules:

Rule 1: The current speaker chooses the next speaker by asking an opinion, question, or request

Rule 2: Another person decides to start speaking

Rule 3: The current speaker continues talking

Adjacency Pairs

- A fundamental concept in the study of conversation analysis
- It refers to pairs of utterances in a conversation
- **Turn-taking**
  - Adjacency pairs consist of an initiating turn (the first utterance) and a responding turn (the second utterance)
  - Turn-taking used to coordinate conversation

    A: Shall we meet at 8:00?
    B: Um, can we meet a bit later?

Examples of Adjacency Pairs

- Question-Answer Pair:
  - A: "What time is the meeting?"
  - B: "It starts at 10 AM.“
- Greeting-Greeting Pair:
  - A: "Good morning!"
  - B: "Morning!“
- Offer-Acceptance/Rejection Pair:
  - A: "Would you like some coffee?"
  - B (Acceptance): "Yes, please."
  - B (Rejection): "No, thank you.“
- Complaint-Apology Pair:
  - A: "You forgot to pick up the dry cleaning."
  - B: "I'm sorry, I completely forgot."

Adjacency Pairs (Cont.)

- Sometimes adjacency pairs get embedded in each other, so it may take some time for a person to get a reply to their initial request or statement

  A: Shall we meet at 8:00?
  B: Wow, look at him?

  A: Yes what a funny hairdo!
  B: Um, can we meet a bit later?

Back-channeling

- Back-channeling refers to the use of verbal and nonverbal signals by listeners to indicate their attention, understanding, or agreement with the speaker.
- One common form of back-channeling is the use of vocalizations such as "uh-huh," "um," "ah," or "mmm."
- Back-chenneling indicates
  - signal to continue: signal that the person is listening
  - following: show that person is following or understanding what the speaker is saying.

Further conversational rules

Farewell rituals

- Bye then, see you, yeah bye, see you later….

Implicit and explicit cues

- For instance, looking at watch or fidgeting with coat and bags
- Explicitly saying,
  - “Well, I have to go now. I got a lot of work to do”
  - “Oh dear, look at the time, I must go, I’m running late…”

Breakdowns in conversation

When someone says something that is misunderstood:

- Speaker will repeat with emphasis:

  A: “This one?”

  B: “No, I meant that one!”

- Also use tokens:

  Eh? Quoi? Huh? What?

What happens in online conversations?

- Do the same conversational rules apply?
- Are there different kinds of breakdowns?
- How do people repair them for:
  - Email?
  - Instant messaging?
  - Texting?
  - Skype or other videoconferencing software?

Activity

- How do people repair breakdowns when conversing via email?
- Repairing the conversation when things are left unsaid or are unclear
  - For example, when someone proposes an ambiguous meeting time, where the date and day given don't match up for the month, the person receiving the message may begin their reply by asking politely,

    "Did you mean this month or June?"

    rather than baldly stating the other person's error, for example,

    "The 13th May is not a Wednesday!"

New social rules during COVID-19

- For how to behave and interact with others when communicating via video conferencing
- Muting yourself when not talking
- Raising a yellow hand when wanting to speak
- Other emoji reactions became commonly used to signify various forms of praise, emotions and non-verbal feedback, e.g.
  - clicking on the red heart and party hat icons that momentarily appear in someone’s window and then disappear after 10 seconds

Online collaboration and communication

- Much research on how to support conversations when people are remote
- Many applications have been developed
  - email, videoconferencing, instant messaging, chatrooms
- To what extent do they mimic or move beyond existing ways of conversing?

Talking with Chatbots

- Conversational User Interfaces, such as chatbots, are more sophisticated in how they emulate turn-taking that takes place in face-to-face conversations
  - e.g. Replika
- Recent developments in AI are changing how we converse and interact with AI
  - e.g, ChatGTP

Talking with Chatbots

Figure: A snippet of my conversation with Replika, a conversational companion, showing turn-taking

Talking with Chatbots

- Many chatbots are designed to answer customer queries, such as answering a question about how to obtain a refund on an online purchase.
- They are trained for a particular set of questions that they can answer well,
- if the customer asks an unexpected question, they are unable to answer other than provide a stock phrase, such as "I am not sure about that."

Dilemma: Is it OK to talk with a dead person using a chatbot?

- Eugenia Kuyda lost a close friend in a car accident who was only in his 20s
- She took all his texts sent over the course of his life and made a chatbot using them
- Chatbot responds to text messages so that Eugenia can talk to her friend as if he was alive
- Is this a creepy or comforting way to deal with grief?
  - Is it respectful of the dead person?

Early videoconferencing: VideoWindow (Bellcore, 1989)

- Shared space that allowed people 50 miles apart to carry on a conversation as if in same room drinking coffee together
- 3 x 8 foot ‘picture-window’ between two sites with video and audio
- People did interact via the window, but strange things happened (Kraut, 1990)

Diagram of VideoWindow in use

Findings of how VideoWindow System was used

- Talked constantly about the system
- Spoke more to other people in the same room rather than in other room
- When trying to get closer to someone in the other place it had opposite effect—participants went out of range of the camera and microphone
- No way of monitoring this

Remote collaboration and communication

- During the pandemic, Zoom and Teams became more widely used.
- From being largely a means to chat with others online, these tools rapidly evolved to be more like virtual meeting rooms, providing a range of other functions that could support online activities.
  - These included letting users choose from a variety of background screens,
  - having a parallel window to chat in,
  - assigning people to breakout rooms for smaller group discussions,
  - and providing a range of emojis intended to be used in the moment as a form of expression

Videoconferencing and telepresence rooms

- Many to choose from to connect multiple people (e.g. Zoom, Teams, Google Meet)
- Customized telepresence rooms for groups

360 degree Camera

- 360 cameras have also started to be used with videoconferencing that capture a panoramic view of a meeting room.
- Instead of having a webcam positioned to face only one direction, the 360 camera is typically placed in the middle of the group meeting on a table.
- This enables remote team members to see all those present during the meeting.

360 degree Camera

Figure - The Meeting Owl setup being used in a hybrid meeting

It shows what participants see

- which is a split screen view of all the members present in the meeting and the person talking blown up beneath that.

https://meetingstore.co.uk/product/meeting-owl-3/

Current videoconferencing

- Teams, Zoom etc., have greatly extended how we communicate while providing tools to make is easier to switch between talking and working together
- Zoom fatigue came into being (Bailenson, 2021)
  - excessive amounts of close-up eye gaze
  - intense cognitive load
  - increased self-evaluation from staring at a video of oneself
  - physically being in the same place for hours on end

Microsoft prototype of a technology-enhanced hybrid meeting

- MS conducted research on how to make hybrid meetings more inclusive
- They experimented with where to place the video feeds of remote members on a shared screen
- Instead of having them appear at the top, they placed them at the bottom
- The effect was to promote better eye gaze, which is at the same level as the participants in the room.
- In the rest of the display, documents are presented that can be annotated and changed in real time, in relation to what the team is working on.

Microsoft prototype of a technology-enhanced hybrid meeting

http://www.microsoft.com/en-us/worklab/designing-the-new-hybrid-meeting-experience

Metaverse

- It is a digital space inhabited by digital representations of people and things (Microsoft, 2021).
- Its earliest incarnation was in the form of virtual worlds, like Second Life, where people could join and create online spaces to socialize, learn, and hang out in.
- The virtual worlds were represented in 3D graphics that were navigated and experienced on a desktop

Metaverse

The metaverse promises to be a space where people can socialize, work and play.

Second Life Metaverse

- Second Life is a virtual world developed by Linden Lab, first launched in 2003.
- It allows users to create and customize avatars, interact with each other, explore virtual landscapes, participate in activities, and even create and trade virtual property and goods.
- a digital universe where users can socialize, engage in commerce, build structures, and participate in various activities.

Metaverse Fashion Week

Dolce & Gabbana cat model showing off a snazzy skin that people can purchase to wear on their own avatar in Second Life

coin3.net/the-first-ever-metaverse-fashion-week-digital-fashion-is-here-to-stay

The Metaverse

- Meta's vision of three friends socializing in a 3D world represented as torso avatars
- Users experience each other through donning VR headsets

Facebook's vision of socializing in a 3D world using VR

- Two avatars talking at a virtual table
- Users experience each other through donning VR headsets

Zuckerberg's vision of the Metaverse

- As shown in the last 2 figures, avatars are enjoying the outdoors, moving their arms and hands with the VR controllers, as if they are together, even though they are apart in the physical world.
- These avatars don't have any legs!
  - The reason for this is largely down to technical limitations.

Telepresence

- A technology that allows individuals to feel as though they are present in a remote location (or “telepresent”), even though they are physically distant.
- Telepresence systems typically use advanced audiovisual and communication technologies to create a sense of immersion and realism, enabling users to interact with people and environments as if they were physically there.

Telepresence

- Robots are often used as a means of enabling physical presence and interaction in remote locations.
  - Telepresence robots, also known as remote presence robots or telepresence avatars, are robotic devices equipped with cameras, microphones, speakers, and displays that allow users to see, hear, and communicate with others in a remote environment in real-time.

Telepresence robots

Enable people to attend events who could not do so, such as by controlling their robot remotely

- In places such as schools, conferences, and museums
- Early example: Beam+
- Often dressed up to appear like the person to others at the event
- Positive experience of being there

Susan Lechelt at ACM CHI

Telepresence and social presence

- *Telepresence* refers to one party being present with another party, who is present in a physical space, such as a meeting room
- *Social presence* refers to the feeling of being there with a real person when in virtual reality

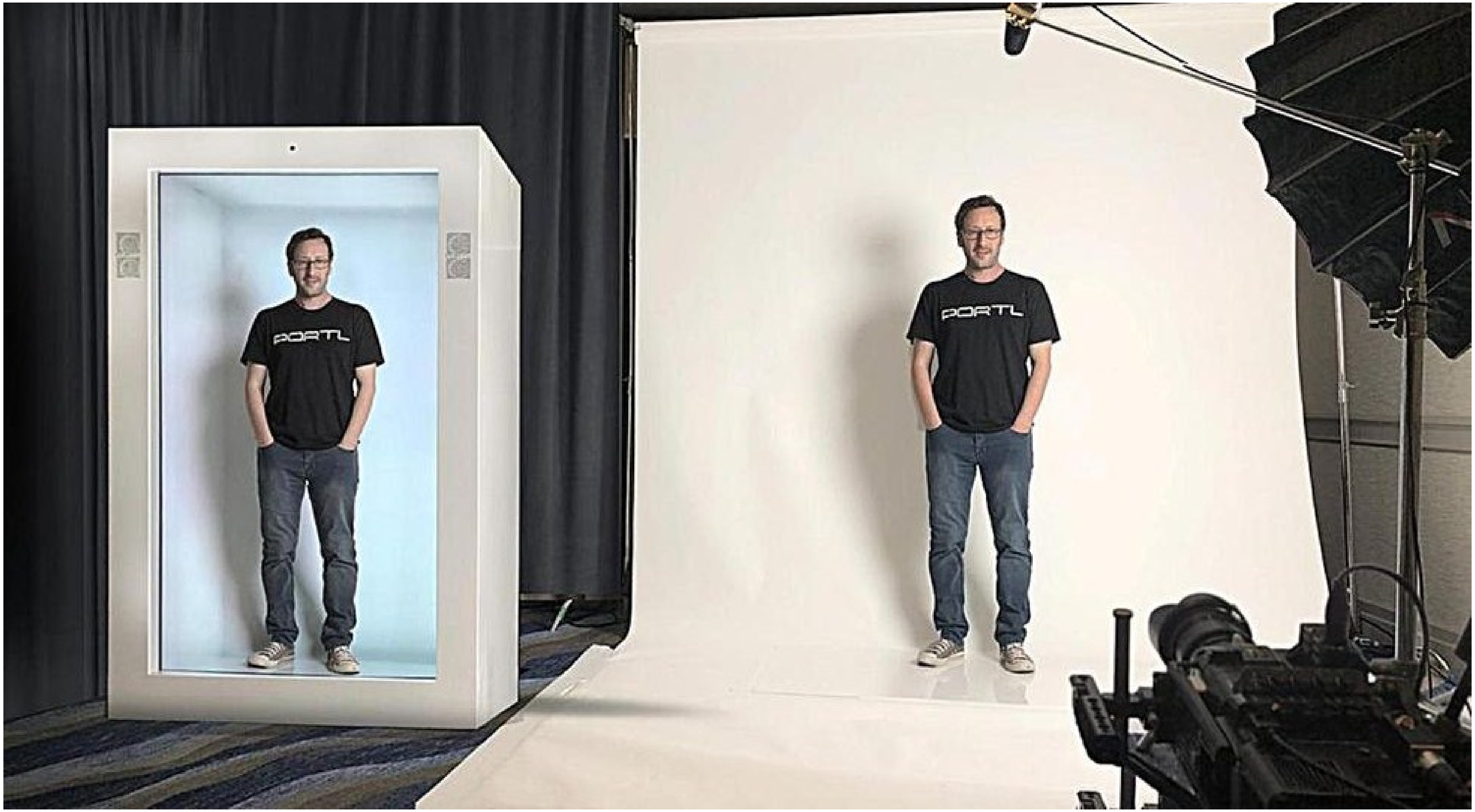
How much realism and immersion are necessary..?

- …in telepresence to make it compelling?
- *Telepresence rooms* try to make remote people appear to be life-like
  - Use multiple high definition cameras with eye-tracking features and directional microphones
- Does FaceTime have as much presence as more high definition settings?

People-in-a-box

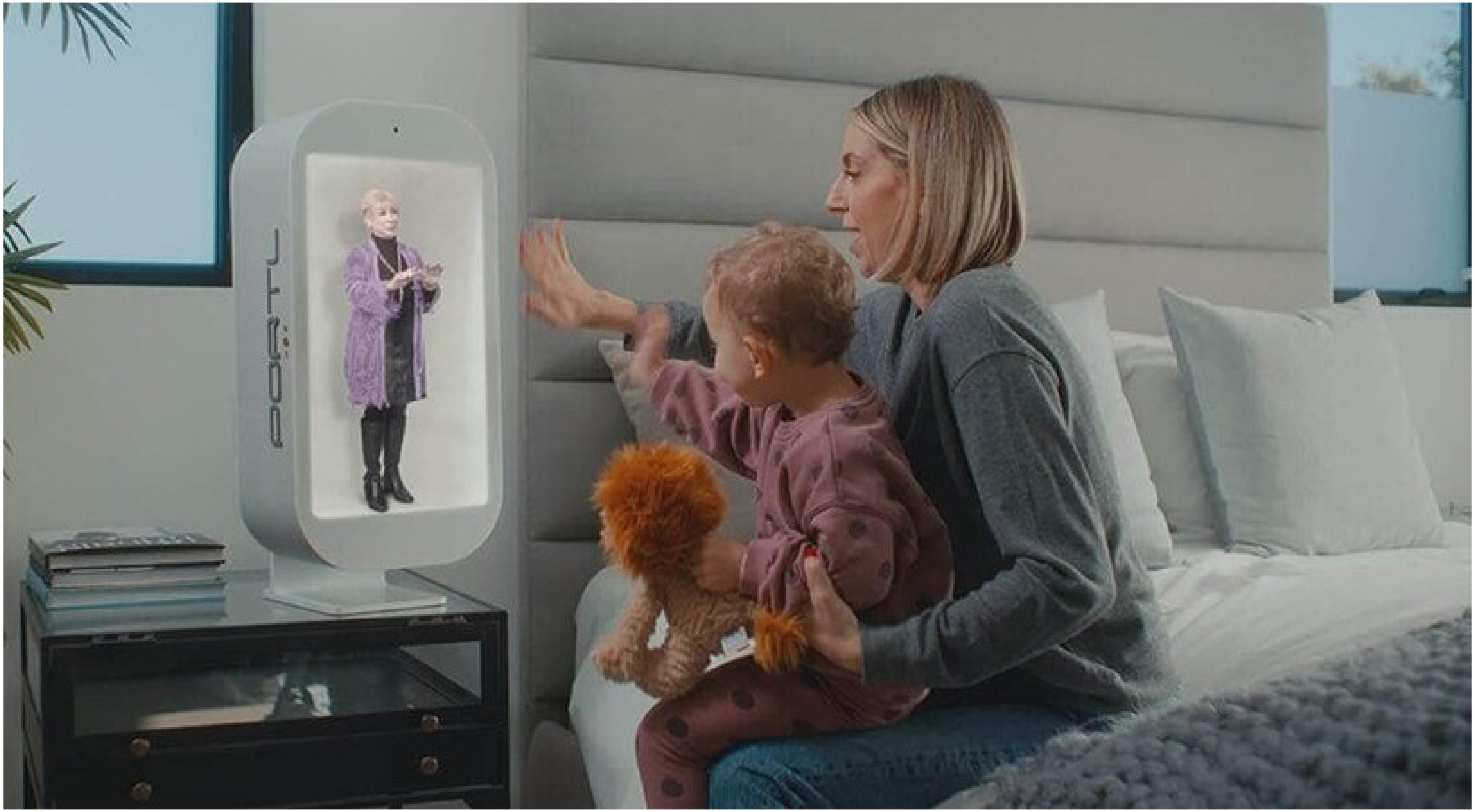
- Instead of talking to a remote family member via videoconferencing, in the future it may be possible to talk to a miniature-size (or life-size) 3D image of them.
- **Proto** (formerly Portl), a startup company set up by David Nussbaum in 2020, has been exploring this possibility through creating boxes where a 3D digital person appears in them.
  - They look so lifelike they could almost be there
  - Even captures the person's shadows as they move around

The future: 3D person in a box?



David Nussbaum demonstrating how they capture and present the Proto person in a box

Proto M

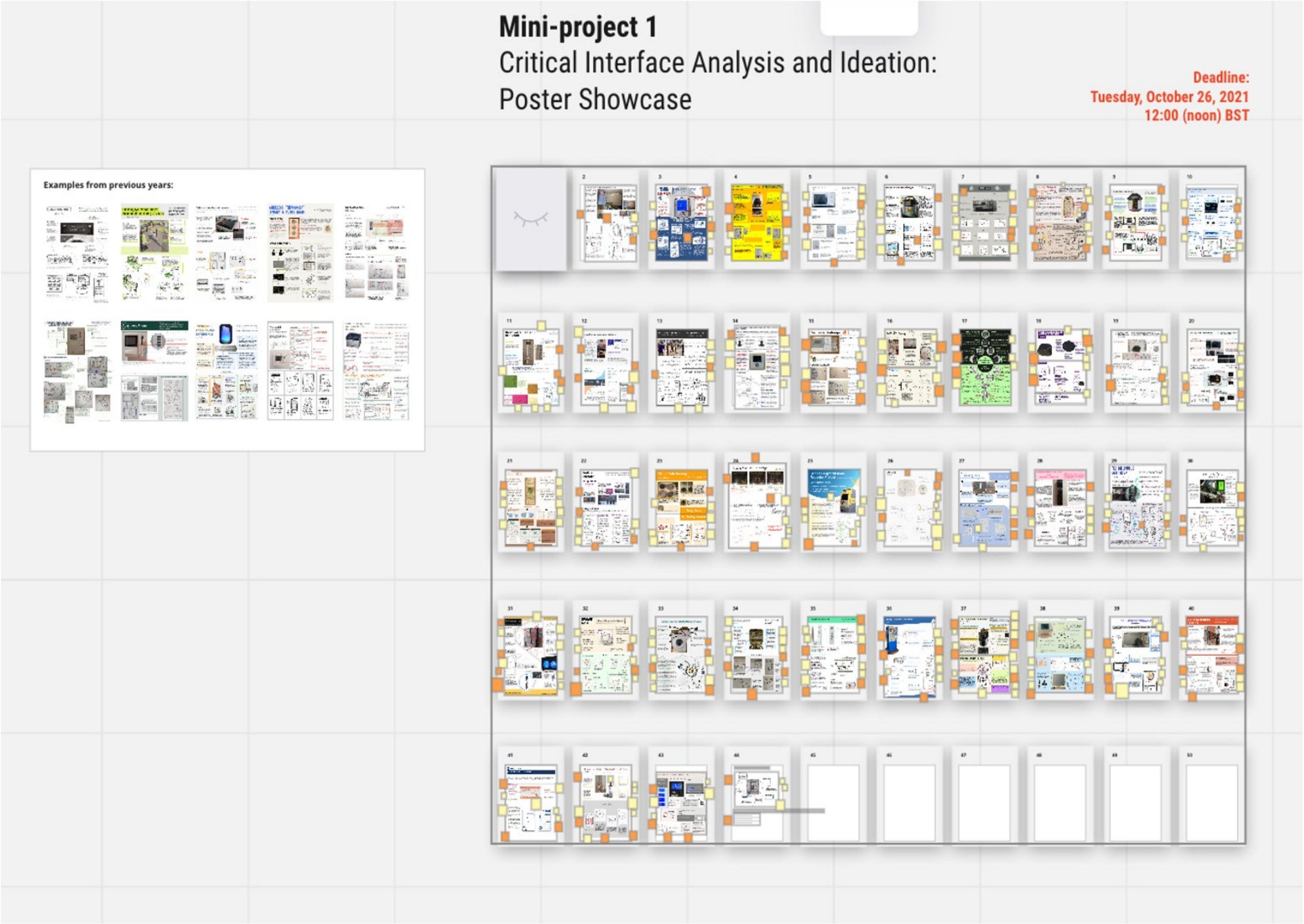


Talking with a 3D video of granny in a box. The embedded camera at the top of the box faces the mother and child so that Granny can see and hear them in real time.

Online collaborative tools

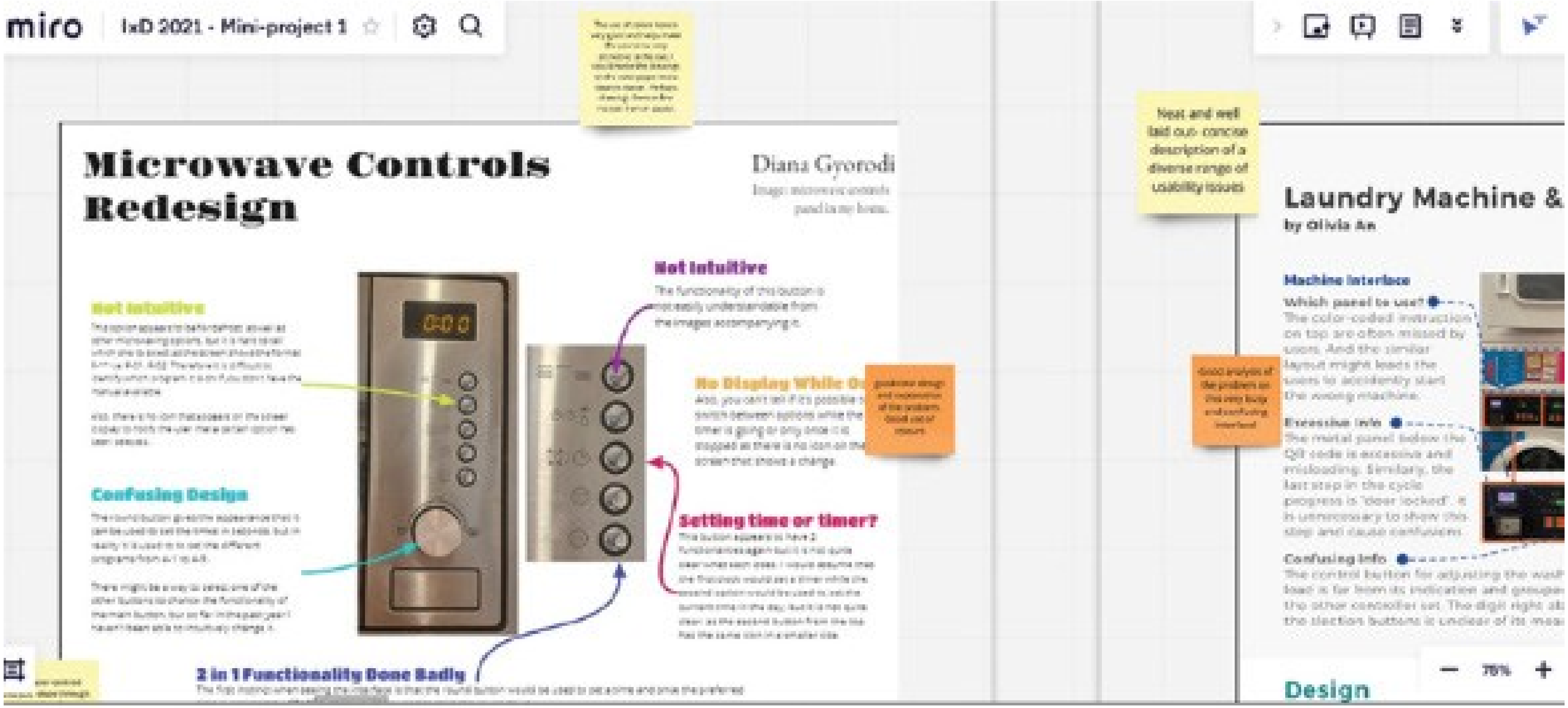
- Now more commonplace in our everyday and working lives
  - e.g. the sharing of calendars, word processing, design and project management tools like Slack
- Places for sharing knowledge
- Tools like Miro, Overleaf and Google Docs enable online collaborative creation and editing of reports, designs, etc.,

Collaborative online tools



A Miro board used in an online class on interaction design where students upload their posters and add comments using yellow post-it notes. The professor and teaching assistants also added theirs using orange ones.

Collaborative online tools



A zoomed-in screen of two of the student posters.

What is co-presence?

- Co-located groups who want to collaborate
- Many technologies have been designed to:
  - Enable groups to work, learn and socialize more effectively together
  - For example, tabletops, whiteboards, and public displays

Co-Presence


Summary
Divided into 6 regions
Costs calculated for
– new buildings
49% of total
– additions
31% of total
– renovations
20% of total
31%

Co-Presence

- To understand how effective these technologies are at supporting co-presence,
  - it is important to consider the **coordination and awareness mechanisms** already in use by people in face-to-face interactions
  - and then to see how these have been adapted or replaced by the technology.

Coordination mechanisms

- When a group of people act or interact together, they need to coordinate themselves
  - For example, when playing football or navigating a ship
- To do so, they use:
  - Verbal and non-verbal communication
  - Schedules, rules, and conventions
  - Shared external representations

In person coordinating mechanisms

- Talk is central
- Non-verbal also used to emphasize and as a substitute
  - e.g., nods, shakes, winks, glances, gestures, and hand-raising
- Formal meetings
  - Explicit structures such as agendas, memos, and minutes are employed to coordinate the activity

Awareness mechanisms

- Involves knowing who is around, what is happening, and who is talking with whom (Dourish and Bly, 1992)
- Peripheral awareness
  - Keeping an eye on things happening in the periphery of vision
    - Being aware of objects, events, or things that are not directly in one's line of sight but are still within the field of vision.
  - Overhearing and overseeing—allows tracking of what others are doing without explicit cues

Awareness mechanisms

- Situational awareness
  - Being aware of what is happening around you in order to understand how information and your actions will affect ongoing and future events
  - the ability to understand and interpret the overall situation or context in which one finds themselves.
    - For example, air traffic control or an operating theatre

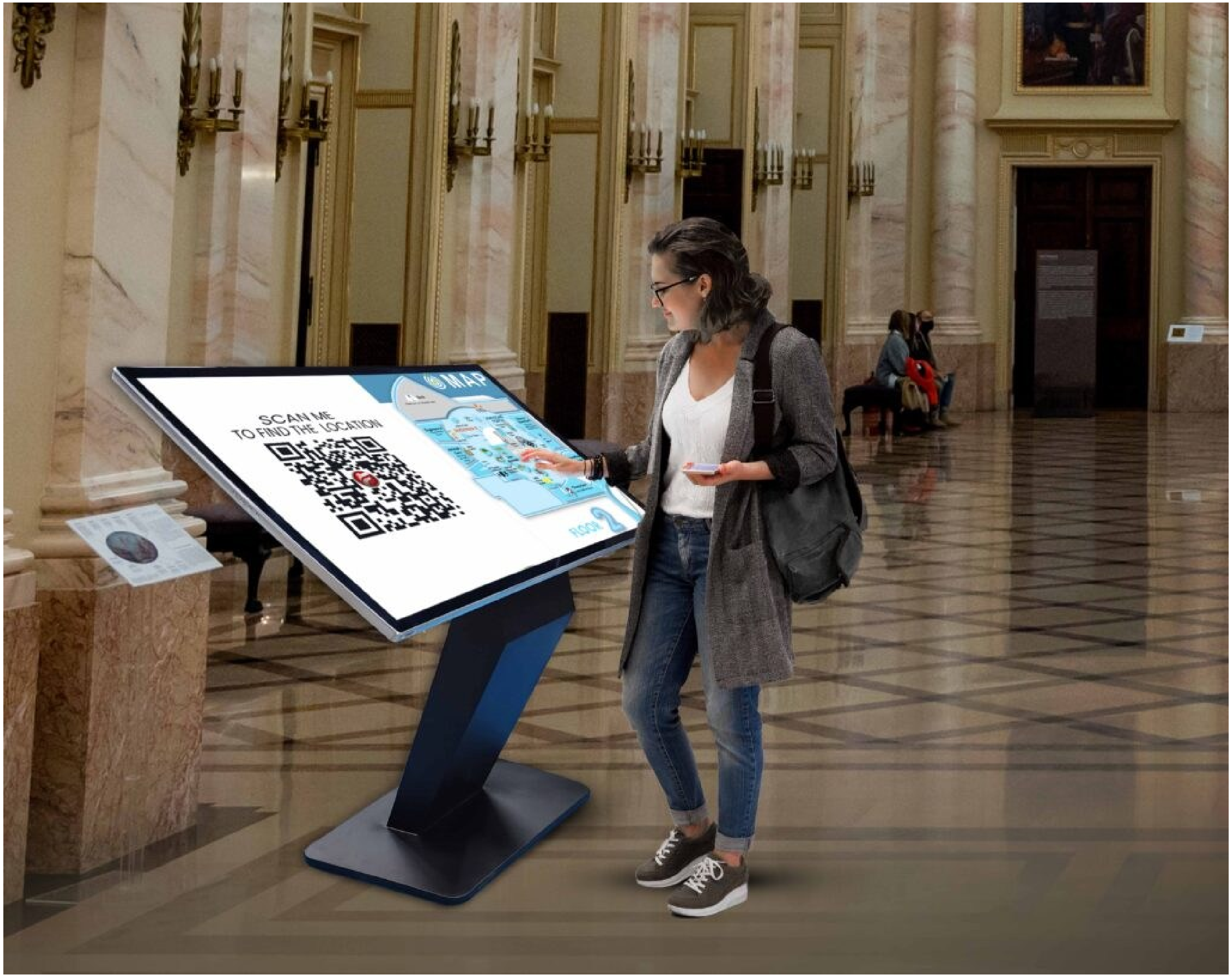
Sharable interfaces

- Technologies designed to capitalize on existing forms of coordination and awareness mechanisms
  - E.g., whiteboards, large touchscreens, and multitouch tables
- Enable groups of people to collaborate while interacting at the same time with content on the surfaces.

Sharable interfaces

- Several studies investigating whether they help people to work together better, have found:
  - More equitable participation
  - More natural to work around
  - More comfortable sitting around a table than standing in front of a display

Public touch display



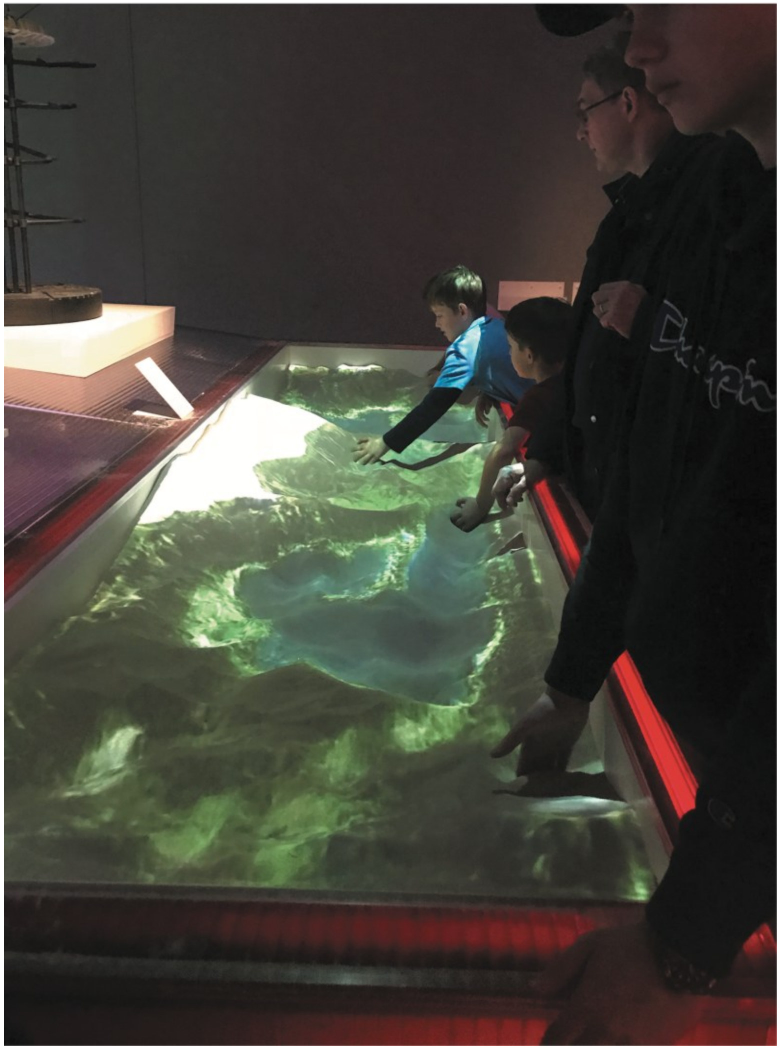
The Reflect Table

- Reflect is a meeting table equipped with an array of 8x16 LEDs and three microphones in its center.
- LEDs lit up to reflect how much each member of the group spoke
- Used microphones in front of each individual to do this
- Study showed those who spoke the most changed their behavior the most
- Those who spoke the least did not change their behavior
- Why do you think this is?

The Reflect Table

Playing together in same space

Visitors creating a landscape together using an augmented reality(AR) sandbox at the V&A Museum in London



Playing together in same space

- Visitors sculpt landscapes out of sand
- System reacts with changing superimposed digital colored landscape
- Enables creative forms of collaboration
- Example: 2 children playing at the sandbox
  - One says, “let’s turn this land into a sea.”
  - The other replies, “OK, but let’s make an island on that.”

PeopleLens: An AR head-mounted device that enhances a blind child's spatial awareness of those around them

Social Games

- Designed to facilitate social interaction
- can be played indoors or outdoors, with and without technology.
  - examples include board games, tabletop games, and videogames (such as Minecraft)
- Each player is aware of other players' presence, their actions and how well they are playing

Social online games

- Can involve creating a community, where competition, collaboration, peer pressure, rebellion, jealousy, and so on are all played out in their various forms
- Matt Richetti (2022) has proposed three heuristics for evaluating them:
  - Does the social game involve synchronous or asynchronous player interaction, where players either chat with each other or they take turns.
  - Is the social interaction symmetrical or asymmetrical, in the sense that does forming a relationship require input from both parties or can they be formed unilaterally by a single party?
  - Does the social relationship involve strong or weak ties in the sense of whether the relationships between players become deep and long lasting or are they transitory?
- By asking these questions, game designers can think about the kinds of social interactions they want to support

Summary

- Social interaction is central to our everyday lives
- Social mechanisms, like turn-taking, enable us to collaborate and coordinate our activities
- Keeping aware of what others are doing and letting others know what you are doing are important aspects of collaborative working and socializing
- Many technologies have been built to support telepresence, social presence, and co-presence
- Social media has brought about significant changes in how people keep in touch and manage their social lives